

An integrated lung disease built from agent-curated datasets

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Abstract

The abundance of nucleotide sequence data has grown into the order of trillions of bases. As public archives grow in scale, so too does their fragmentation and complexity, meaning that disease-focused meta-analyses remain difficult to accomplish. We identified the need for producing atlas-scale datasets that both incorporate work spanning studies and are uniformly processed and targeted to disease areas. Here, we develop a reproducible pipeline for building a lung disease-specific, integrated, and study-aware atlas from a broad, AI-curated atlas of single-cell RNA sequencing (scRNA-seq) data. We present an, analysis-ready atlas that successfully integrates XXX,XXX,XXX cells across X,XXX files and XXX distinct scientific projects, preserving cell-type structure and reducing batch effects.

Introduction

The scale and heterogeneity of single-cell RNA sequencing (scRNA-seq) datasets are rapidly expanding (Luecken et al., 2022; Mueller et al., 2026). Multiple scientific efforts have been undertaken in integrating collections of scRNA-seq datasets with improvements in sequencing technology and analysis methods (Rozenblatt-Rosen et al., 2017). The Sequence Read Archive (SRA) is the largest repository of scRNA-seq data (Youngblut et al., 2025) and is complex by nature because a wide array of laboratories and organizations might contribute to it (Leinonen et al., 2011). The Gene Expression Omnibus (GEO) is also community-derived while enforcing reporting standards relating to data and metadata (Clough & Barrett, 2016). Other initiatives, such as the Human Cell Atlas (Rozenblatt-Rosen et al., 2017) and CZ CELLxGENE Discover (CZI Cell Science Program et al., 2025), aim to balance scalability with accommodation for efficient meta-analysis, by combining standardization of cell matrices and extensive metadata with community driven contribution (CZI Cell Science Program et al., 2025). The recent development of scBaseCount has attempted to address the need for consistent processing in these aggregated collections of scRNA-seq data by leveraging agentic AI-driven metadata unification to uniformly process scRNA-seq datasets in the resource (Youngblut et al., 2025).

Materials and methods

Building an integrated lung atlas using scBaseCount data required developing robust single-cell workflows in Python using Scanpy (Wolf et al., 2018). We enforced observability through extensive logging configured with the Python logger.

Data acquisition

scBaseCount data, which includes expression matrices, are hosted on Google Cloud Platform in Google Cloud Storage (GCS) buckets (Youngblut et al., 2025). We used the 2026-01-12 snapshot of scBaseCount. Before downloading any matrices, we identified a subset of the available matrices that represented lung tissue. We accomplished this by filtering scBaseCount’s tissue metadata using a regular expression containing lung-like words, such as “lung,” “pulmonary,” “alveolus/i/ar.” We sequentially fetched datasets whose metadata passed that filter from GCS and computed their MD5 hashes on disk. Due to the extensive cumulative size of the datasets, we uploaded the datasets themselves, along with the newly computed hashes, to an internally owned and maintained Cloudflare R2 bucket, serving as a production mirror of scBaseCount, allowing us to remove local files from disk.

Atlas construction & quality control

To integrate the acquired individual datasets as a single atlas-scale **h5ad** file, we built a pipeline to sequentially download scBaseCount datasets from the R2 mirror, filter them by performing quality control operations, determine if what remained of the filtered datasets was sufficient for inclusion in the atlas, align the datasets to a fixed gene axis, and concatenate the datasets to one in-memory atlas. We wrote the concatenated atlas to disk on pipeline completion.

To ensure datasets matched scBaseCount, we computed dataset MD5 hashes following download from R2 and compared them the MD5 hashes stored as metadata alongside the corresponding R2 objects, the latter having been computed upon upload. We enforced identity between the upload and download hashes.

Next, we filtered cells based on tunable parameters. We did not drop genes for rarity. We flagged genes as likely mitochondrial, ribosomal, and hemoglobin genes by applying regular expressions to the gene names, and computed per-cell quality metrics. We first dropped cells with a total gene count below 200 genes. We dropped cells whose mitochondrial and ribosomal gene counts were at least 20% and 50%, respectively, of total gene counts. We did not filter cells on hemoglobin gene counts. Following these filters, we applied a cutoff on number of remaining cells and percentage of cells remaining relative to the initial number of cells in the dataset. We did not include datasets in the atlas if they had fewer than 100 cells remaining or if fewer than 50% of cells remained in the dataset following filtering. We also excluded datasets in which every cell lacked a usable CZ CELLxGENE **cell_type** annotation (CZI Cell Science Program et al., 2025). We configured every mentioned threshold as a tunable parameter. To ensure that the values we settled on were well calibrated, we recorded the number of cells dropped by each filter, allowing us to rerun the filtering pipeline with different filtering parameters. We did not apply doublet detection. We chose filtering parameters based on recommendations proposed by Luecken and Theis (2019) and enriched them with the ribosomal filter and hemoglobin flagging. Before we concatenated the filtered datasets along the cell axis to the atlas **h5ad** file, we reindexed each dataset onto the scBaseCount **geneInfo.tab** gene list, discarding genes absent from that list and filling missing reference genes with zeros, to ensure concatenated matrices used a shared gene axis. Since Youngblut et al. (2025) already indexed scBaseCount matrices against a shared STAR reference, this reindex did not change the genome reference.

Embedding & batch correction

We developed a method to embed and cluster the atlas using the CZ CELLxGENE **cell_type** key present in the datasets. First, we normalized gene counts per cell so that each cell had the same total gene count.

Then, we logarithmized the data matrix by the natural logarithm, selected the top 2,000 highly variable genes (HVG) stratified by BioSample—a number in the range of recommended HVGs according to Luecken and Theis (2019)—and z -scaled the expression of each gene. We then applied principal component analysis (PCA), computing the first 50 PCs. We then recorded the minimum number of PCs in the range of 15–50 explained at least 50% of the data’s cumulative variance. We applied the processing steps through PCA using Scanpy (Wolf et al., 2018).

Following PCA, we applied Harmony to the PCs to correct for one batch effect (Korsunsky et al., 2019). The scBaseCount snapshot from which we collected data did not contain any keys coarser than SRA experiment accession, which was effectively a one-to-one file key. We identified the National Center for Biotechnology Information’s (NCBI) BioProject accession as a key to use for applying batch correction to the atlas. BioProject accessions group SRA experiment accessions by project type, project scope, technical methodology, or other groupings (Barrett et al., 2012). To collect the BioProject accession for a given SRA experiment accession, along with other relevant metadata, we queried the European Nucleotide Archive’s (ENA) portal API (Amid et al., 2020). As a result of the International Nucleotide Sequence Database Collaboration (INSDC), ENA and NCBI both host BioProject accessions (Arita et al., 2021).

To visualize and cluster the data, we computed the nearest neighbors distance matrix and a neighborhood graph of observations and applied Scanpy’s reproducible, single threaded UMAP dimensionality reduction implementation (McInnes et al., 2018). We clustered the neighborhood graph using the Leiden algorithm (Traag et al., 2019). To select an optimal Leiden clustering resolution, we sequentially applied Leiden clustering to the neighborhood graph at a range of resolutions from 0.1–2.0 at intervals of 0.1, computed the Jaccard index (also known as “intersection over union” or “IoU”) of each CZ CELLxGENE `cell_type`-Leiden cluster pair, and determined the optimal matching using SciPy’s linear sum assignment implementation (Virtanen et al., 2020). We recorded the cost of the assignment at each Leiden clustering resolution, and retained the Leiden clustering resolution with the lowest cost as the canonical Leiden clustering resolution. TODO: decide on if/how to include RF.

To test our clustering methodology, we implemented it as a set of importable functions and applied them to datasets corresponding with single SRA experiment accessions before applying it to the atlas. We took a stratified sample of 100,000 cells, where strata are BioProjects and counts per stratum are proportional to each BioProject’s share of the full atlas (with at least one cell per BioProject). We benchmarked the batch key using scIB (Luecken et al., 2022).

Results

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The atlas contained X,XXX datasets from X,XXX eligible datasets, spanning XXX BioProject accessions. After filtering, the atlas contained X,XXX,XXX barcodes, down from X,XXX,XXX barcodes (Table 1). The atlas contained XX CZ CELLxGENE cell_type labels (Fig 1).

Table 1. Atlas contained X,XXX datasets (X,XXX,XXX barcodes) across XXX BioProject accessions, of X,XXX eligible datasets (X,XXX,XXX barcodes) following filtering.

Filter	Datasets dropped, <i>n</i> (%)	Barcodes dropped, <i>n</i> (%)
MD5 hash (mismatch)	0 (0)	X,XXX (X)
CZ CELLxGENE cell_type (labels absent)	XXX (X)	XX,XXX (X)
Min. genes per cell (< 200)	—	X,XXX (X)
Max. mitochondrial fraction (≥ 20%)	—	XXX (X)
Max. ribosomal fraction (≥ 50%)	—	XX (X)
Min. cells remaining (< 100)	XX (X)	XX,XXX (X)
Min. fraction of cells remaining (< 50%)	XX (X)	XX,XXX (X)

A total of the XXX datasets were dropped. Of those, 0 were dropped due to MD5 hash mismatch, XXX (X% total datasets) were dropped due to fully absent CZ CELLxGENE cell_type labels. Of the XX,XXX barcodes that were dropped, X,XXX (X% total barcodes) were dropped for gene counts of less than 200, XXX (X% total barcodes) were dropped for having 20% or more of total gene count being mitochondrial, and XX (X% total barcodes) for having 50% or more of total gene counts being ribosomal. Following filtering, datasets dropped for excessive barcode dropout. XX (X% total datasets) were dropped for retaining fewer than 100 cells, and XX (X% total datasets) were dropped for retaining less than 50% of cells following filtering.

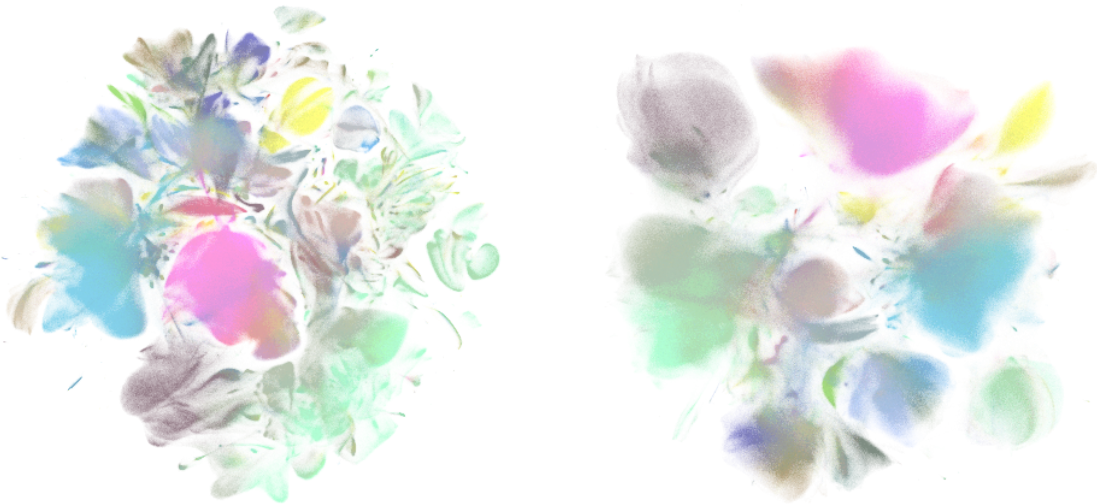


Fig 1. Single-file integrated lung disease atlas derived from scBaseCount. UMAP of the integrated atlas colored by CZ CELLxGENE cell_type. A: Uncorrected embedding. B: Harmony-corrected embedding (batch key: BioProject).

On atlas scale, batch effect removal per CZ CELLxGENE `cell_type` label, measured by the k-nearest-neighbor batch effect test (kBET) proposed by Büttner et al. (2019) and implemented by Luecken et al. (2022) was lower from the uncorrected PCs compared to the batch corrected PCs, the latter using NCBI BioSample as batch key (TODO:decide whether to cite all scib metrics in text). The aggregate biological conservation score proposed by Luecken et al. (2022) decreased from 0.53 to 0.51 as a result of batch correction, while the batch correction aggregate score increased from 0.28 to 0.45 (Fig 2).

Method	Bio conservation					Batch correction					Aggregate score		
	Isolated labels	KMeans NMI	KMeans ARI	Silhouette label	cUSI	BRAS	iLUSI	KBET	Graph connectivity comparison	PCR	Batch correction	Bio conservation	Total
X_pca_harmony	0.45	0.46	0.20	0.44	0.99	0.75	0.02	0.48	0.47	0.52	0.45	0.51	0.49
X_pca	0.50	0.51	0.21	0.45	0.99	0.50	0.00	0.31	0.60	0.00	0.28	0.53	0.43

Fig 2. Batch correction metrics before and after Harmonization of PCs. Benchmark results on a stratified atlas subset comparing uncorrected PCs and Harmony-corrected PCs (batch key: NCBI BioSample). Aggregate biological conservation decreased from 0.53 to 0.51; aggregate batch correction increased from 0.28 to 0.45. kBET was lower for uncorrected than batch-corrected embeddings.

Sweeping leiden

Discussion

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Conclusion

Supporting information

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